

PYTHON DATA TYPES

Multiple Choice Questions

Basic Data Types

1. Which of the following is an integer value?
 A) 3.14 B) 5 C) 2+3j D) "5"
2. Which of the following is a floating-point number?
 A) 10 B) 3.5 C) "3.5" D) True
3. Which represents a complex number in Python?
 A) 5 B) 3.14 C) 2+3j D) "2+3j"
4. Which of the following is a Boolean value?
 A) 0 B) 1 C) True D) "True"
5. Which of the following is a string?
 A) 'Hello' B) 10 C) 2.5 D) True
6. Which data type is used to store multiple values in order and is mutable?
 A) Tuple B) List C) String D) Dictionary
7. Which of the following is immutable?
 A) List B) Dictionary C) Tuple D) Set
8. Which symbol is used to create a list?
 A) () B) [] C) { } D) < >
9. Which symbol is used to create a tuple?
 A) () B) [] C) { } D) " "
10. Which of the following is a dictionary?
 A) [1,2,3] B) (1,2,3) C) {"a":1} D) "abc"

Understanding Data Types

11. A dictionary stores data in _____ pairs.
 A) Index-Value B) Key-Value C) Value-Value D) Key-Index
12. Which type can store mixed data values?
 A) int B) List C) float D) bool
13. Which data type is ordered?
 A) Dictionary B) List C) Set D) NoneType

14. What is the type of None?
A) int B) String C) NoneType D) bool
15. Which of the following is immutable?
A) String B) List C) Dictionary D) Set
16. Which of the following is mutable?
A) Tuple B) String C) List D) int
17. Which data type can be used as a dictionary key?
A) List B) Tuple C) Dictionary D) Set
18. Which of the following is NOT a sequence type?
A) List B) Tuple C) Dictionary D) String
19. Which is a valid Boolean constant in Python?
A) TRUE B) False C) false D) TRUE()
20. Which data type represents absence of a value?
A) 0 B) False C) None D) ""
21. Which data type is most suitable for storing fixed data that should not change?
A) List B) Tuple C) Dictionary D) String
22. Why is a list preferred over a tuple in many applications?
A) Faster B) Mutable C) Smaller D) Immutable
23. Which data type is best for key-based access of data?
A) List B) Tuple C) Dictionary D) String
24. Which type is safest when data should remain unchanged?
A) List B) Tuple C) Dictionary D) Set
25. Which data type is best for storing a student record with name, roll and marks?
A) List B) Dictionary C) Tuple D) String
26. Which data type allows duplicate values?
A) Dictionary Keys B) List C) Set D) NoneType
27. Which data type is generally more memory efficient?
A) List B) Tuple C) Dictionary D) Set
28. Which data type maintains the order of inserted elements?
A) Dictionary B) List C) Set D) NoneType
29. Which data type is most suitable for mathematical calculations?
A) int / float B) String C) List D) Tuple

30. Which is best for representing “no result” or “missing value”?

- A) 0 B) False C) None D) “”

Advanced MCQs

31. Which of the following is a valid complex number?

- A) $2+3i$ B) $2+3j$ C) $2+j3$ D) $j+3$

32. Which data type is used to store textual data?

- A) String B) Integer C) Float D) Complex

33. Which data type supports indexing and slicing?

- A) String B) List C) Tuple D) All of these

34. Which data type is called a mapping type?

- A) List B) Tuple C) Dictionary D) String

35. Which of the following is mutable?

- A) String B) Tuple C) List D) Integer

36. Which of the following can contain duplicate elements?

- A) Set B) List C) Dictionary Keys D) NoneType

37. Which data type can store heterogeneous values?

- A) List B) Tuple C) Dictionary D) All of these

38. Which of the following is not mutable?

- A) List B) Dictionary C) Tuple D) Set

39. Which built-in function returns the type of a variable?

- A) `datatype()` B) `class()` C) `typeof()` D) `type()`

40. Which data type is represented by curly braces with key-value pairs?

- A) Set B) Tuple C) Dictionary D) List

Match the Following

41. Match the Following

Column A

Column B

- | | |
|-------------|---------------------|
| (a) int | (i) Whole Number |
| (b) float | (ii) Decimal Number |
| (c) complex | (iii) $a+bj$ |
| (d) bool | (iv) True/False |

- A) a-i, b-ii, c-iii, d-iv B) a-ii, b-i, c-iv, d-iii C) a-iii, b-iv, c-i, d-ii D) a-iv, b-iii, c-ii, d-i

42. Match the Following**Column A****Column B**

- | | |
|----------------|-----------------|
| (a) List | (i) Mutable |
| (b) Tuple | (ii) Immutable |
| (c) String | (iii) Immutable |
| (d) Dictionary | (iv) Mutable |

- A) a-i, b-ii, c-iii, d-iv B) a-ii, b-i, c-iv, d-iii C) a-iii, b-iv, c-i, d-ii D) a-iv, b-iii, c-ii, d-i

43. Match the Following**Column A****Column B**

- | | |
|---------|------------------|
| (a) [] | (i) List |
| (b) () | (ii) Tuple |
| (c) { } | (iii) Dictionary |
| (d) " " | (iv) String |

- A) a-i, b-ii, c-iii, d-iv B) a-ii, b-i, c-iv, d-iii C) a-iii, b-iv, c-i, d-ii D) a-iv, b-iii, c-ii, d-i

44. Match the Following**Column A****Column B**

- | | |
|----------|---------------|
| (a) None | (i) No Value |
| (b) True | (ii) Boolean |
| (c) 5 | (iii) Integer |
| (d) 2.5 | (iv) Float |

- A) a-i, b-ii, c-iii, d-iv B) a-ii, b-i, c-iv, d-iii C) a-iii, b-iv, c-i, d-ii D) a-iv, b-iii, c-ii, d-i

45. Match the Following**Column A****Column B**

- | | |
|----------------|------------------------|
| (a) Dictionary | (i) Key-Value |
| (b) List | (ii) Ordered |
| (c) Tuple | (iii) Fixed Collection |
| (d) String | (iv) Text |

- A) a-i, b-ii, c-iii, d-iv B) a-ii, b-i, c-iv, d-iii C) a-iii, b-iv, c-i, d-ii D) a-iv, b-iii, c-ii, d-i

Assertion-Reasoning**46. Assertion (A): Tuples are immutable.**

Reason (R): Elements of a tuple cannot be modified after creation.

- A) Both A and R are true and R explains A
 B) Both A and R are true but R does not explain A
 C) A is true but R is false
 D) A is false but R is true

47. **Assertion (A):** Lists are mutable.
Reason (R): Elements can be added, removed or modified in a list.
- A) Both A and R are true and R explains A
 B) Both A and R are true but R does not explain A
 C) A is true but R is false
 D) A is false but R is true
48. **Assertion (A):** Dictionary data is stored using keys.
Reason (R): Values are accessed using corresponding keys.
- A) Both A and R are true and R explains A
 B) Both A and R are true but R does not explain A
 C) A is true but R is false
 D) A is false but R is true
49. **Assertion (A):** None represents the absence of a value.
Reason (R): None belongs to the NoneType data type.
- A) Both A and R are true and R explains A
 B) Both A and R are true but R does not explain A
 C) A is true but R is false
 D) A is false but R is true
50. **Assertion (A):** A tuple can be used as a dictionary key.
Reason (R): Tuples are immutable.
- A) Both A and R are true and R explains A
 B) Both A and R are true but R does not explain A
 C) A is true but R is false
 D) A is false but R is true

Fill in the Blanks

Choose the correct option.

51. Whole numbers are stored using the _____ type.
- A) float B) string C) int D) bool
52. Decimal numbers are stored using the _____ type.
- A) int B) float C) complex D) string
53. Complex numbers use _____ as the imaginary unit.
- A) i B) I C) j D) k

54. True and False belong to the _____ type.
 A) string B) bool C) int D) float
55. A sequence of characters is called a _____.
 A) tuple B) list C) string D) dictionary
56. A _____ is a mutable sequence.
 A) tuple B) list C) string D) bool
57. A _____ is an immutable sequence.
 A) list B) dictionary C) tuple D) set
58. A dictionary stores data in _____ pairs.
 A) index-value B) value-value C) key-value D) key-index
59. _____ represents the absence of a value.
 A) False B) 0 C) None D) ""
60. Data types that can be modified are called _____.
 A) immutable B) mutable C) sequence D) numeric

Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	B	11	B	21	B	31	B	41	A	51	int
2	B	12	B	22	B	32	A	42	A	52	float
3	C	13	B	23	C	33	D	43	A	53	j
4	C	14	C	24	B	34	C	44	A	54	bool
5	A	15	A	25	B	35	C	45	A	55	string
6	B	16	C	26	B	36	B	46	A	56	list
7	C	17	B	27	B	37	D	47	A	57	tuple
8	B	18	C	28	B	38	C	48	A	58	key-value
9	A	19	B	29	A	39	D	49	B	59	None
10	C	20	C	30	C	40	C	50	A	60	mutable